

Begonnen am	Donnerstag, 11. Juli 2024, 10:20
Status	Beendet
Beendet am	Donnerstag, 11. Juli 2024, 11:00
Verbrauchte Zeit	40 Minuten 21 Sekunden
Bewertung	52,35 von 100,00

Frage 1

Teilweise richtig

Erreichte Punkte 4,00 von 10,00

In the lecture we discussed diffusion models where $x_0 \sim p(x)$ is our observed data and $x_{1:T}$ are the latent variables. The "forward process" model was defined as

$$q(x_t|x_{t-1}) = N(x_{t-1}\sqrt{\alpha}, (1-\alpha)I).$$

where N is the normal distribution, I is the identity matrix and 0 does not depend on t . Using the properties of the normal distribution, $q(x_t|x_0)$ can be calculated analytically. Give the correct form of $q(x_t|x_0)$:

- a) $N(\sqrt{\alpha^t}x_0, (1-\alpha)I)$
- b) $N(\sqrt{\alpha}x_0, (1-\alpha^t)I)$
- c) $N(\sqrt{\alpha^t}x_0, (1-\alpha^t)I)$
- d) $N(\sqrt{\alpha^t}x_0, (1-\alpha)^tI)$

☒ c) 

The ELBO for T time steps is given by

$$\underbrace{E_{q(x_1|x_0)}[\log p_\theta(x_0 | x_1)]}_{L_1} - \underbrace{D_{\text{KL}}(q(x_T | x_0) || p(x_T))}_{L_2} - \underbrace{\sum_{t=2}^T \mathbb{E}_{q(x_t|x_0)}[D_{\text{KL}}(q(x_{t-1} | x_t, x_0) || p_\theta(x_{t-1} | x_t))]}_{L_3}.$$

Tick the correct answers about the terms L_1, L_2, L_3 :

- ☐ L_1 is the reconstruction loss measuring the log probability of the original data given the sampled gaussian noise.
- ☒ L_1 is the reconstruction loss measuring the log probability of the original data given the first latent step.
- ☒
- ☐ L_2 measures how well the backward process produces the prior, i.e., the original data, from a noisy latent and approaches 0 for low T (low noise).
- ☐ L_1 is the reconstruction loss measuring the log probability of the first denoising step given the sampled gaussian noise.
- ☐ L_2 measures how well the forward process produces the prior, i.e., a standard gaussian, from the original data and approaches 0 for high T (only noise).
- ☐ L_3 is minimized when the approximate noise step matches the ground-truth noise step.
- ☐ L_3 is minimized when the approximate denoising step matches the ground-truth denoising step
- ☐ L_3 is efficiently computed using simple monte-carlo integration. It converges quickly because we integrate over gaussians.
- ☐ L_3 is efficiently computed using the closed form Kullback-Leibler divergence between gaussians.

Frage 2

Teilweise richtig

Erreichte Punkte 6,67 von 10,00

Consider a hierarchical VAE with 2 levels, meaning we have two latent variables z_1 and z_2 . The generative process is straight forward, first sample z_2 , then sample z_1 given z_2 and finally sample x given z_1 . We want utilize variational inference with a family of variational posteriors $Q(z_1, z_2 | x)$.

How does the posterior factorize?

- a) $Q(z_1, z_2 | x) = q(z_1 | x)q(z_2 | z_1)$
- b) $Q(z_1, z_2 | x) = q(z_2 | x)q(z_1 | z_2)$
- c) $Q(z_1, z_2 | x) = q(z_2 | x)q(z_2 | z_1)$
- d) $Q(z_1, z_2 | x) = q(z_1 | x)q(z_1 | z_2)$

a) ☒ ✓

What is the ELBO of this 2 level VAE?

- a) $ELBO(x) = E[\log(p(x|z_1)) - KL(q(z_1|x)||p(z_1|z_2)) - KL(q(z_1|z_2)|p(z_1))]$
- b) $ELBO(x) = E[\log(p(x|z_2)) - KL(q(z_2|x)||p(z_2|z_1)) - KL(q(z_1|z_2)|p(z_1))]$
- c) $ELBO(x) = E[\log(p(x|z_2)) - KL(q(z_2|x)||p(z_2|z_1)) - KL(q(z_1|z_2)|p(z_2))]$
- d) $ELBO(x) = E[\log(p(x|z_1)) - KL(q(z_1|x)||p(z_1|z_2)) - KL(q(z_2|z_1)|p(z_2))]$

d) ☒ ✓

Training VAEs can be challenging.

Mark the correct answer:

Sampling from the latent is tricky during training. ☒ ✗

Frage 3

Teilweise richtig

Erreichte Punkte 3,35 von 10,00

A random sample X_1, X_2, X_3 of size 3 is drawn iid. from a population with mean $w = 3$ and variance $\sigma^2 = 2$.

Consider the estimators $\hat{w}_1 = \frac{1}{3}(X_1 + X_2 + X_3) + 1$ and $\hat{w}_2 = \frac{1}{3}(X_1 + 2X_2 + X_3)$ of the mean w .

a) Which of the following statements about the estimators' biases is correct?

w_2 is biased, w_1 is unbiased. ☒ ✗

b) Compute the variance of the two estimators:

$\text{Var}(\hat{w}_1) =$ ☒ ✓

$\text{Var}(\hat{w}_2) =$ ☒ ✗

c) Based on the MSE, which estimator would you choose?

w_1 ☒ ✓

IMPORTANT: Only choose an answer if you are sure that it is correct. If you choose no answer, no points will be deducted. If you choose a wrong answer, it will cost you as many points as you would have gotten with the right answer.

Frage 4

Teilweise richtig

Erreichte Punkte 6,67 von 10,00

Let $w, x \in \mathbb{R}$. Let X be a random variable distributed according to a "supernice" parametrized distribution $p(x, w)$, so that p satisfies all the regularity conditions for any theorem we discussed in the lecture. X shall satisfy $\text{Var}(X) = \frac{1}{w+1}$ and by $I_p(w)$ we denote the associated Fisher information. Moreover, let the following estimators be given:

- An estimator $\hat{w}$ of w so that $E(\hat{w}) = w$.
- For an iid. sample of size l the associated Maximum Likelihood estimator (MLE) $\hat{w}_{ML}(l)$.
- The estimator $\hat{w}_{ML}(\infty) = \lim_{l \rightarrow \infty} \hat{w}_{ML}(l)$

Tick the statements that are true for **any** choice of the terms defined above. In case there are statements about limits, the precise notion of convergence does not matter.

Wählen Sie eine oder mehrere Antworten:

- ☒ a. $\text{Var}(\hat{w}) \geq \frac{1}{I_p(w)}$. ✓
- ☐ b. $\text{Var}(X) \geq I_p(w)$.
- ☐ c. $\hat{w}_{ML}(l)$ is unbiased for any l .
- ☐ d. The MLE for $E(X)$ for a sample of size l can be computed as $\frac{1}{\hat{w}_{ML}(l)+1}$.
- ☐ e. The variances of $\hat{w}$ and $\hat{w}_{ML}(\infty)$ can be bounded from below simultaneously by the Cramer Rao lower bound.
- ☒ f. $\hat{w}_{ML}(\infty)$ is efficient and unbiased. ✓

Frage 5

Teilweise richtig

Erreichte Punkte 2,50 von 5,00

Which statements about Neural ODEs are true:

Wählen Sie eine oder mehrere Antworten:

- ☒ a. In Neural ODEs, the update $x_{l+1} = x_l + f(x_l, \theta)$, where l corresponds to the respective layer output, is treated as Euler discretization of $\frac{dx}{dt} = f(x, t, \theta)$. ✓
- ☐ b. Neural ODEs apply a forward pass: $x(t + \Delta t) = \int_t^{t+\Delta t} f(x(\tau), \tau, \theta) d\tau$ where the time t takes the role of the layer l .
- ☐ c. We use an implicit ODE solver to ensure the same time steps in the forward and in the backward ODE.
- ☐ d. For the backward ODE, the loss serves as initial condition.

Frage 6

Teilweise richtig

Erreichte Punkte 2,50 von 5,00

Describe the Kuramoto-Shivashinsky (KS) equation:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + \frac{\partial^2 u}{\partial x^2} + \frac{\partial^4 u}{\partial x^4} = 0$$

Tick **all** correct statements.

Wählen Sie eine oder mehrere Antworten:

- ☐ a. The KS equation is a time-dependent PDE.
- ☒ b. The KS equation is a homogeneous equation. ✓
- ☒ c. The KS equation is a fourth-order equation. ✓
- ☐ d. The KS equation is a non-linear equation.

Frage 7

Teilweise richtig

Erreichte Punkte 2,50 von 10,00

For probabilistic PCA, we assume all marginal and conditional distributions to be Gaussian:

- Gaussian prior distribution $p(z) = \mathcal{N}(0, \mathbf{I})$, where $z \in \mathbb{R}^h$
- Gaussian conditional distribution $p(x|z) = \mathcal{N}(Wz + \mu, \sigma^2 \mathbf{I})$ for the observed data $x_1, \dots, x_l \in \mathbb{R}^k$, where $\mu \in \mathbb{R}^k$ and $\mathbf{I} \in \mathbb{R}^{k \times k}$.

Tick **all** statements that are true.

Wählen Sie eine oder mehrere Antworten:

- ☐ a. The matrix $W \in \mathbb{R}^{k \times h}$ maps from the h -dimensional latent space to the k -dimensional data space.
- ☐ b. To generate data we assume a deterministic dependency $x = Wz + \mu$.
- ☐ c. The maximum of the log-likelihood of $\ln(p(x)) = \sum_{i=1}^l \ln(p(x_i|\mu, WW^T + \sigma^2 \mathbf{I}))$ with respect to $W \in \mathbb{R}^{k \times h}$ is obtained when the h eigenvectors of the data covariance matrix are chosen to be those whose eigenvalues are the smallest.
- ☐ d. The maximum of the log-likelihood of $\ln(p(x)) = \sum_{i=1}^l \ln(p(x_i|\mu, WW^T + \sigma^2 \mathbf{I}))$ with respect to μ yields $\mu = \frac{1}{l} \sum_{i=1}^l x_i = \bar{x}$.
- ☐ e. The true posterior $p(z|x)$ is intractable. Nevertheless, due to the properties of the Gaussians, we can estimate it.
- ☐ f. pPCA is the simplest non-linear generative model.
- ☒ g. The expectation $\mathbb{E}_{p(x|z)}(x)$ yields μ . ✓
- ☐ h. For data generation, we choose a value for the latent z , and then sample x from an isotropic Gaussian distribution with mean $Wz + \mu$ and covariance $\sigma^2 \mathbf{I}$.

Frage 8

Teilweise richtig

Erreichte Punkte 6,67 von 10,00

Let us consider the standard statistical learning theory (SLT) setting from the lecture, i.e.

- Data points $\mathbf{Z} = (x_i, y_i)_{i=1}^l$ sampled iid. from $p(x, y)$ supported in $X \times \{-1, 1\}$
- A model class $\mathcal{G}$ of functions $g : X \rightarrow \{-1, 1\}$.
- The zero-one loss function L_{z_0} .
- For $g \in \mathcal{G}$ and L_{z_0} the associated generalization error (risk) $R(g)$ and the associated empirical error $R_{\text{emp}}(g, l)$.
- $g_{\mathcal{G}} = \arg \min_{g \in \mathcal{G}} R(g)$: the best model in the model class.
- g_l : an algorithm depending on the given data.

Tick **all** correct statements related to this setting.

Wählen Sie eine oder mehrere Antworten:

- ☒ a. For a fixed $g \in \mathcal{G}$, the convergence $R_{\text{emp}}(g, l)$ to $R(g)$ for $l \rightarrow \infty$ can be ensured by the law of large numbers. ✓
- ☐ b. For any $g \in \mathcal{G}$ and $\varepsilon > 0$, Hoeffding's inequality gives an upper bound on $p(|R_{\text{emp}}(g, l) - R(g)| \geq \varepsilon)$, that decays at most polynomially with respect to l .
- ☒ c. The notion of consistency allows for statements about convergence of $R(g_l)$ to $R(g_{\mathcal{G}})$ for $l \rightarrow \infty$. ✓
- ☐ d. A crucial fact about SLT is, that you always assume some explicit knowledge about the data distribution in order to create your algorithms.
- ☐ e. The main purpose of the concept of uniform convergence is to ensure that $R(g)$ and $R_{\text{emp}}(g, l)$ are close for any $g \in \mathcal{G}$ and large l .
- ☐ f. Consistency always implies uniform convergence. The converse, however, is not true in general.

Frage 9

Teilweise richtig

Erreichte Punkte 2,50 von 10,00

Let us consider the standard statistical learning theory (SLT) setting from the lecture, i.e.

- Data points $\mathbf{Z} = (x_i, y_i)_{i=1}^l$ sampled iid. from $p(x, y)$ supported in $X \times \{-1, 1\}$
- A model class $\mathcal{G}$ of functions $g : X \rightarrow \{-1, 1\}$.
- The zero one loss function L_{z_0} .
- For $g \in \mathcal{G}$ and L_{z_0} the associated generalization error (risk) $R(g)$ and the associated empirical error $R_{\text{emp}}(g, l)$.
- $\mathcal{N}(\mathcal{G}, l)$: the shattering coefficient of $\mathcal{G}$ for a sample of size l .

Tick the correct statements about generalization bounds in this setting. In case there are statements about limits, you do not need to care about the precise notion of convergence.

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Let $g \in \mathcal{G}$. The generalization bounds related to the setting described above are of the form $R(g) \leq R_{\text{emp}}(g, l) + \text{capacity term} + \text{confidence term}$. The capacity term measures the size of $\mathcal{G}$, whereas the confidence term accounts for uncertainty. ✓
- ☐ b. Any sample of size l has more than $\mathcal{N}(\mathcal{G}, l)$ realizations/functions in $\mathcal{G}$.
- ☐ c. Let $g \in \mathcal{G}$. The symmetrization lemma allows to upper bound $|R(g) - R_{\text{emp}}(g, l)|$ by a difference $|R_{\text{emp}}(g, l) - R'_{\text{emp}}(g, l)|$ of empirical risks for two iid. samples of sizes l .
- ☐ d. The principle of symmetrization allows to incorporate the shattering coefficient in the capacity term of the generalization bounds.
- ☐ e. For finite VC-dimension $d \leq l$, $\mathcal{N}(\mathcal{G}, l)$ can be upper bounded by a term that is polynomial in l .

Frage 10

Teilweise richtig

Erreichte Punkte 5,00 von 10,00

In Variational Autoencoders (VAEs), we maximize the evidence lower bound (ELBO), which is given by

$\text{ELBO}(x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \mathbb{D}_{\text{KL}}(q_\phi(z|x)||p(z))$. Here, $q_\phi(z|x)$ is the encoder network which maps the data x into a latent space z , and $p_\theta(x|z)$ is the decoder network which maps from the latent space z back to the data.

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Maximizing the ELBO means maximizing $\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)]$ and minimizing $\mathbb{D}_{\text{KL}}(q_\phi(z|x)||p(z))$. ✓
- ☐ b. We maximize $\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)]$ by maximizing the mean-squared error of the reconstructed data $p_\theta(x|z)$ and the ground truth data x .
- ☐ c. Maximizing $\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)]$ is similar to maximum likelihood where we want to find the model parameter to minimize the likelihood of observing the image.
- ☒ d. We minimize $\mathbb{D}_{\text{KL}}(q_\phi(z|x)||p(z))$ by minimizing the KL divergence between the encoder distribution $q_\phi(z|x)$ and the latent distribution $p(z)$. ✓
- ☐ e. In a perfectly trained VAE, the ELBO equals the log-likelihood of the data.
- ☐ f. In VAEs, we don't have access to the distribution $p(z|x)$, thus we estimate this distribution with $q_\phi(z|x)$. This is called variational inference.
- ☐ g. We need the reparametrization trick since the computed VAE loss is based on a forward pass that includes a stochastic sampling procedure. If we use the reparametrization trick $z = \mu + \sigma\mathbf{I} + \epsilon$, the forward pass becomes fully deterministic.
- ☐ h. The encoder is not involved in data generation.

Frage 11

Richtig

Erreichte Punkte 10,00 von 10,00

Consider a binary classification task in the 2-dimensional plane with labels $+1$ and -1 , respectively. Your model class $\mathcal{G}$ consists of all origin centered circles, i.e. for given radii $r > 0$, the model $g_r \in \mathcal{G}$ assigns the label $+1$ to all points which have distance less than r from the origin and -1 otherwise. By d we denote the corresponding VC-dimension of $\mathcal{G}$.

Tick the **correct** statements.

Hint: drawing some of the examples might help.

Wählen Sie eine oder mehrere Antworten:

- ☐ a. For the one point configuration consisting of $\mathbf{x}_1 = (0, 0)$ the label $y_1 = -1$ can not be realized. Thus we can conclude $d < 1$.
- ☐ b. For the two point configuration consisting of $\mathbf{x}_1 = (1, 0)$ and $\mathbf{x}_2 = (2, 0)$ the labelling $y_1 = +1$ and $y_2 = -1$ can be realized. Therefore $d \geq 2$.
- ☐ c. For the two point configuration consisting of $\mathbf{x}_1 = (1, 0)$ and $\mathbf{x}_2 = (0, 1)$, the labelling $y_1 = -1$ and $y_2 = +1$ can not be realized. Thus we can conclude $d < 2$.
- ☒ d. For any two points $\mathbf{x}_1$ and $\mathbf{x}_2$ denote their distances to the origin by r_1 and r_2 , respectively. Without loss of generality assume $r_1 \leq r_2$. Then the labelling $y_1 = -1$ and $y_2 = +1$ can not be realized, which ensures that $d < 2$. ✓